

ZIMIEX TABLET 10mg

DESCRIPTION

White to off white capsule shaped tablets debossed with “713” on one side and plain on the other side.
Dimension: About 8 mm x 4 mm

COMPOSITION

Each tablet contains 10mg ezetimibe.

INDICATIONS

Primary Hypercholesterolaemia

ZIMIEX administered alone, or with an HMG-CoA reductase inhibitor (statin), is indicated as adjunctive therapy to diet in patients with primary (heterozygous familial and non-familial) hypercholesterolaemia.
ZIMIEX, administered in combination with fenofibrate, is indicated as adjunctive therapy to diet for the reduction of elevated total-C, LDL-C, Apo B, and non-HDL-C in patients with mixed hyperlipidemia.

Prevention of Cardiovascular Events

ZIMIEX, administered with a statin, is indicated to reduce the risk of cardiovascular events (cardiovascular death, nonfatal myocardial infarction, nonfatal stroke, hospitalization for unstable angina, or need for revascularization), in patients with coronary heart disease (CHD) and a history of acute coronary syndrome (ACS).

Homozygous Familial Hypercholesterolaemia (HoFH)

ZIMIEX, administered with a statin, is indicated for patients with HoFH. Patients may also receive adjunctive treatments (e.g., LDL apheresis).

Homozygous Sitosterolaemia (Phytosterolaemia)

ZIMIEX is indicated for the reduction of elevated sitosterol and campesterol levels in patients with homozygous familial sitosterolaemia.

DOSAGE AND ADMINISTRATION

The patient should be on an appropriate lipid-lowering diet and should continue on this diet during treatment with ZIMIEX. Route of administration is oral.

Use in Patients with Primary Hypercholesterolemia

The recommended dose of ZIMIEX is 10 mg once daily, used alone, with a statin or with fenofibrate. ZIMIEX can be administered at any time of the day, with or without food.
ZIMIEX may be administered with a statin (in patients with primary hypercholesterolemia) or with fenofibrate (in patients with mixed hyperlipidemia) for incremental effect. For convenience, the daily dose of ZIMIEX may be taken at the same time as the statin or fenofibrate, according to the dosing recommendations for the respective medications.

If ZIMIEX 10mg tablets are used in combination with a statin therapy, the dosage instructions for that particular statin should be consulted.

When initiating lipid lowering treatment, which includes ZIMIEX 10mg Tablets and a statin in combination, the indicated usual initial dose of that particular statin should be used or the already established higher statin dose should be continued.

If the statin dose is to be increased for the first time or further, the dosage instructions of that particular statin should be followed (such as dose increase only after at least 4 weeks of regular use of the combination without any change).

The stepwise increase of the statin dose in combination treatment results in a relatively small additional decrease of LDL-C, but increases the risk of dose-related adverse events of the statin. This has to be considered for the risk-benefit-assessment when the statin dose is considered.

Use in Patients with Coronary Heart Disease

Combination Therapy with a Statin

For incremental cardiovascular event reduction in patients with coronary heart disease, ZIMIEX 10 mg may be administered with a statin with proven cardiovascular benefit.

Use in the Elderly

No dosage adjustment is required for elderly patients. However, greater sensitivity of some older individuals cannot be ruled out.

Use in Pediatric Patients

Children and adolescents ≥ 10 years: No dosage adjustment is required.

Children < 10 years: Treatment with ZIMIEX is not recommended.

Use in Hepatic Impairment

No dosage adjustment is required in patients with mild hepatic insufficiency (Child Pugh score 5 to 6). Treatment with ZIMIEX is not recommended in patients with moderate (Child Pugh score 7 to 9) or severe (Child Pugh score > 9) liver dysfunction (see WARNINGS AND PRECAUTIONS).

Use in Renal Impairment

No dosage adjustment is required for renally impaired patients.

Co-administration with bile acid sequestrants

Dosing of ZIMIEX should occur either ≥ 2 hours before or ≥ 4 hours after administration of a bile acid sequestrant.

Instruction for use of the peel-off blisters:

1. Do not crush the tablet

In order to prevent crushing the tablet, do not push against the tablet pocket (Figure A).



Fig. A.

2. Tear off one tablet pocket

The tablet pockets on the blister strip are separated from each other by perforations. Tear off one tablet pocket along the dotted lines (Figure 1).

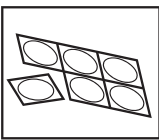


Fig. 1.

3. Peel off the lid

Carefully peel off the lidding foil, starting in the corner where the lidding foil is not sealed (Figures 2 and 3).

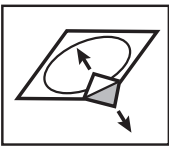


Fig. 2.

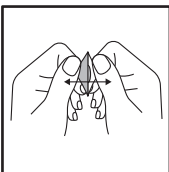


Fig. 3.

4. Take the tablet out of the pocket

CONTRAINDICATIONS

Patients with known hypersensitivity to any component of ZIMIEX TABLET 10mg.

The combination of ezetimibe with a statin is contraindicated in patients with active liver disease or unexplained persistent elevations in serum transaminases.

All statins are contraindicated in pregnant and nursing women. When ZIMIEX is administered with a statin in a woman of childbearing potential, refer to the pregnancy category and product labeling for the statin.

WARNINGS AND PRECAUTIONS

When ezetimibe is to be administered with a statin or with fenofibrate, please refer to the information for that particular statin.

Liver Enzymes

In patients receiving ezetimibe with a statin, consecutive transaminase elevations [≥3x the upper limit of normal (ULN)] have been observed. When ezetimibe is co-administered with a statin, liver function tests should be performed at initiation of therapy and according to the recommendations of the statin.

Hepatic Insufficiency

Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe hepatic insufficiency, ezetimibe is not recommended in these patients.

Fibrates

The co-administration of ezetimibe with fibrates other than fenofibrate has not been studied. Therefore, co-administration of ezetimibe and fibrates (other than fenofibrate) is not recommended.

Fenofibrate

If cholelithiasis is suspected in a patient receiving ezetimibe and fenofibrate, gallbladder studies are indicated and alternative lipid-lowering therapy should be considered.

Cyclosporine

Caution should be exercised when initiating ezetimibe in the setting of cyclosporine. Cyclosporine concentrations should be monitored in patients receiving ezetimibe and cyclosporine.

Skeletal Muscle

Myopathy and rhabdomyolysis are known adverse reactions to HMG-CoA reductase inhibitors and other lipid-lowering drugs.

Anticoagulants

If ezetimibe is added to warfarin, another coumarin anticoagulant, or fludionone, the International Normalised Ratio (INR) should be appropriately monitored.

DRUG INTERACTIONS

It has been shown that ezetimibe does not induce cytochrome P-450 drug metabolizing enzymes. No clinically significant pharmacokinetic interactions have been observed between ezetimibe and drugs known to be metabolized by cytochromes P-450 1A2, 2D6, 2C8, 2C9, and 3A4, or N-acetyltransferase. Ezetimibe had no effect on the pharmacokinetics of dapsone, dextromethorphan, digoxin, oral contraceptives (ethinyl estradiol and levonorgestrel), glipizide, tolbutamide or midazolam during co-administration. Cimetidine, co-administered with ezetimibe, had no effect on the bioavailability of ezetimibe.

Antacids

Concomitant antacid administration decreased the rate of absorption of ezetimibe but had no effect on the bioavailability of ezetimibe. This decreased rate of absorption is not considered clinically significant.

Cholestyramine

Concomitant cholestyramine administration decreased the mean AUC of total ezetimibe (ezetimibe + ezetimibe glucuronide) approximately 55%. The incremental LDL-C reduction due to adding ezetimibe to cholestyramine may be lessened by this interaction.

Cyclosporine

Ezetimibe used in post-renal transplant patients with creatinine clearance of >50 mL/min on a stable dose of cyclosporine, will result in an increase in the mean AUC for total ezetimibe.

250693_s1 - P: Ezetimibe (Zimiex) MY, Tablets 10mg, ALL

GENERAL INFORMATION	NON-PRINTING COLOURS	COLOURS/PLATES
Dimensions: 190x380 mm	1.	1. Black ●
Supplier: Actavis Ltd. MALTA	2.	2.
Min pt size*: 10 pt	3.	3.
		4.
		5.
		6.

Fibrates

Fibrates may increase cholesterol excretion into the bile, leading to cholelithiasis. Therefore, co-administration of ezetimibe with fibrates (other than fenofibrate) is not recommended.

Fenofibrate

Concomitant fenofibrate administration increased total ezetimibe concentrations, which is not considered clinically significant.

Gemfibrozil

Concomitant gemfibrozil administration increased total ezetimibe concentrations, which is not considered clinically significant.

Statins

No clinically significant pharmacokinetic interactions were seen when ezetimibe was co-administered with atorvastatin, simvastatin, pravastatin, lovastatin, fluvastatin or rosuvastatin.

Anticoagulants

Concomitant administration of ezetimibe (10 mg once daily) had no significant effect on bioavailability of warfarin and prothrombin time.

Incompatibilities

Not applicable

USE IN PREGNANCY AND LACTATION

Use in pregnancy

There are no adequate and well-controlled studies of ezetimibe in pregnant women. Ezetimibe should be used during pregnancy only if the potential benefit justifies the risk to the fetus. When ezetimibe is to be administered with a statin, please refer to the package insert for that particular statin.

Use in lactation

It is not known whether ezetimibe is excreted into human breast milk, therefore, ezetimibe should not be used in nursing mothers unless the potential benefit justifies the potential risk to the infant.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINES

No studies on the effects on the ability to drive and use machines have been performed. However, when driving vehicles or operating machines, it should be taken into account that dizziness has been reported.

ADVERSE EFFECTS

Ezetimibe was generally well tolerated; adverse reactions were usually mild and transient. The following are some common drug-related adverse reactions:

Ezetimibe monotherapy or co-administered with a statin		
System organ class	Adverse reactions	Frequency
Blood and Lymphatic System Disorders	thrombocytopenia	Not known
Immune System Disorders	hypersensitivity; including rash; urticaria; anaphylaxis and angio-oedema	Not known
Metabolism and Nutrition Disorders	decreased appetite	Uncommon
Psychiatric Disorders	depression	Not known
Nervous System Disorders	headache	Common
	paraesthesia	Uncommon
	dizziness	Not known
Vascular Disorders	hot flush; hypertension	Uncommon
Respiratory, Thoracic and Mediastinal Disorders	cough	Uncommon
	dyspnoea	Not known
Gastrointestinal Disorders	abdominal pain; diarrhoea; flatulence	Common
	dyspepsia; gastrooesophageal reflux disease; nausea; dry mouth; gastritis	Uncommon
	Pancreatitis; constipation	Not known
	hepatitis; cholelithiasis; cholecystitis	Not known
Skin And Subcutaneous Tissue Disorders	pruritus; rash; urticaria	Uncommon
	erythema multiforme	Not known
Musculoskeletal And Connective Tissue Disorders	myalgia	Common
	arthralgia; muscle spasms; neck pain; back pain; muscular weakness; pain in extremity	Uncommon
	myopathy/rhabdomyolysis	Not known
	fatigue	Common
General Disorders And Administration Site Condition	chest pain, pain; asthenia; oedema peripheral	Uncommon
	ALT and/or AST increased	Common
Investigations	blood CPK increased;	Uncommon
	gamma-glutamyltransferase increased; liver function test abnormal	

Ezetimibe co-administered with fenofibrate

Gastrointestinal disorders: abdominal pain (common)

OVERDOSE AND TREATMENT

A few cases of overdosage with ezetimibe have been reported; most have not been associated with adverse experiences. Reported adverse experiences have not been serious. In the event of an overdose, symptomatic and supportive measures should be employed.

PHARMACOLOGY

Pharmacotherapeutic group: Other lipid modifying agents. ATC code: C10A X09

Ezetimibe is orally active, and has a mechanism of action that differs from other classes of cholesterol-reducing compounds (e.g. statins, bile acid sequestrants [resins], fibric acid derivatives, and plant stanols). The molecular target of ezetimibe is the sterol

transporter, Niemann-Pick C1-Like 1 (NPC1L1), which is responsible for the intestinal uptake of cholesterol and phytosterols. Ezetimibe localises at the brush border of the small intestine and inhibits the absorption of cholesterol, leading to a decrease in the delivery of intestinal cholesterol to the liver. This causes a reduction in hepatic cholesterol stores and an increase in clearance of cholesterol from the blood. Ezetimibe does not increase bile acid excretion (like bile acid sequestrants) and does not inhibit cholesterol synthesis in the liver (like statins).

By inhibiting the absorption of intestinal cholesterol, ezetimibe reduces the delivery of cholesterol to the liver. Statins reduce cholesterol synthesis in the liver. Together these distinct mechanisms provide complementary cholesterol reduction. Ezetimibe, administered with a statin, reduces total-C, LDL-C, Apo B and TG and increases HDL-C in patients with hypercholesterolemia, beyond either treatment alone.

Administration of ezetimibe with fenofibrate is effective in improving serum total-C, LDL-C, Apo B, TG, HDL-C, and non-HDL-C in patients with mixed hyperlipidemia.

PHARMACOKINETICS

Absorption: After oral administration, ezetimibe is rapidly absorbed and extensively conjugated to a pharmacologically active phenolic glucuronide (ezetimibe glucuronide). Mean maximum plasma concentrations (C_{max}) occur within 1 to 2 hours for ezetimibe-glucuronide and 4 to 12 hours for ezetimibe. The absolute bioavailability of ezetimibe cannot be determined as the compound is virtually insoluble in aqueous media suitable for injection. Concomitant food administration (high fat or non-fat meals) had no effect on the oral bioavailability of ezetimibe when administered as Ezetimibe 10-mg tablets. Ezetimibe can be administered with or without food.

Distribution: Ezetimibe and ezetimibe-glucuronide are bound 99.7% and 88 to 92% to human plasma proteins, respectively.

Metabolism: Ezetimibe is metabolised primarily in the small intestine and liver via glucuronide conjugation (a phase II reaction) with subsequent biliary excretion. Minimal oxidative metabolism (a phase I reaction) has been observed in all species evaluated. Ezetimibe and ezetimibe-glucuronide are the major drug-derived compounds detected in plasma, constituting approximately 10 to 20 % and 80 to 90 % of the total drug in plasma, respectively. Both ezetimibe and ezetimibe-glucuronide are slowly eliminated from plasma with evidence of significant enterohepatic recycling. The half-life for ezetimibe and ezetimibe-glucuronide is approximately 22 hours.

Elimination: Following oral administration, total ezetimibe accounted for approximately 93% of the total radioactivity in plasma. Approximately 78% and 11% of the administered radioactivity were recovered in the faeces and urine, respectively, over a 10-day collection period. After 48 hours, there were no detectable levels of radioactivity in the plasma.

Special populations:

Paediatric patients

The absorption and metabolism of ezetimibe are similar between children and adolescents (10-18 years) and adults. Based on total ezetimibe, there are no pharmacokinetic differences between adolescents and adults. Pharmacokinetic data in the paediatric population <10 years of age are not available. Clinical experience in paediatric and adolescent patients (9-17 years) has been limited to patients with HoFH or sitosterolaemia.

Geriatric patients

Plasma concentrations for total ezetimibe are about 2-fold higher in the elderly (≥65 years) than in the young (18 to 45 years). LDL-C reduction and safety profile are comparable between elderly and young subjects treated with ezetimibe. Therefore, no dosage adjustment is necessary in the elderly.

Hepatic insufficiency

No dosage adjustment is necessary for patients with mild hepatic insufficiency. Due to the unknown effects of the increased exposure to ezetimibe in patients with moderate or severe (Child Pugh score >9) hepatic insufficiency, ezetimibe is not recommended in these patients.

Renal insufficiency

No dosage adjustment is necessary for renally impaired patients.

Gender

Plasma concentrations for total ezetimibe are slightly higher (< 20%) in women than in men. LDL-C reduction and safety profile are comparable between men and women treated with ezetimibe. Therefore, no dosage adjustment is necessary on the basis of gender.

Race

Based on a meta-analysis of pharmacokinetic studies, there were no pharmacokinetic differences between Blacks and Caucasians.

STORAGE:

Store below 30°C. Store in original package.

AVAILABILITY

10mg: Blister pack of 3x10's and 10x10's.

For further information, please consult your physician or pharmacist.

Manufactured by:

Actavis Ltd.

BLB 016, Bulebel Industrial Estate, Zetjun ZTN3000, Malta

DATE OF REVISION: May-2025



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